

Genetic factors interact with environmental factors: Type 2 diabetes can have different causes, with different pathways that impact glucose regulation. Tailoring your diabetes prevention (or reversal) efforts to fit your genetic susceptibility may be more effective. For example, people with melatonin receptor variants are at a higher risk of diabetes when eating later at night. You can **prioritize changes** based on your genetic variants.

Adding up the genetic risk:

Researchers have discovered many different genes that increase the relative risk of diabetes. The genetic variants listed below increase (or decrease) the risk of diabetes in most population groups. These variants are well-studied with many replication studies.

For the most part, genetic variants increase the relative risk of diabetes a little bit, often only increasing relative risk by 10-40%. Thus, a single variant may not impact your risk all that much. Instead, the cumulative effects of multiple risk variants seem to be key. Research shows us that, in many cases, **the risk is additive**. In other words, the more risk alleles you carry, the higher the risk.[\[ref\]](#)[\[ref\]](#)

Diet combines with Genetic Risk Score:

Research shows that a 'Western Diet' is associated with an increased risk of diabetes only in people with several genetic risk variants. The Western dietary pattern did not have much of an impact on the risk of diabetes in people without the risk variants.[\[ref\]](#)

Genetic Variants Linked to Diabetes:

Below are some of the most well-researched genetic variants linked to diabetes. If your genotype matches the risk allele, read through the details below the chart for each variant.

Gene	RS ID	Risk Allele	Your Genotype	Notes about the Risk Allele:
MTNR1B	rs10830963	G	CC	Increased risk of diabetes; [ref] increased fasting glucose levels (Melatonin receptor impacts circadian regulation of insulin)
SLC30A8	rs13266634	C	CT	Increased risk of T2D, zinc-related variant [ref]
IRS1	rs2943641	C	TC	Slightly higher risk for diabetes; insulin receptor variant [ref]
HHEX	rs7923837	G	GG	Increases risk of impaired glucose-stimulated insulin response [ref] [ref] [ref]
HHEX	rs1111875	C	CC	
KCNJ11	rs5219	T	TC	Decreased insulin response to glucose [ref] [ref]
KCNQ1	rs2237892	T	CC	Decreased risk of diabetes (good!) [ref] [ref]
PPARG	rs1801282	G	CC	Increased risk of metabolic syndrome and insulin resistance [ref]
PCSK9	rs11591147	T	GG	Decreased LDL cholesterol but an increased risk of diabetes. [ref]
CDKAL1	rs7754840	C	GG	Increased risk of diabetes [ref] [ref] [ref] [ref] [ref]
CDKAL1	rs7756992	G	AA	Increased risk of diabetes [ref] [ref]
CDKAL1	rs10811661	C	TT	TT: Decreased risk of diabetes (good!) [ref]
CDKAL1	rs564398	C	TT	CC: Decreased risk of diabetes (good!) [ref]
TCF7L2	rs7903146	T	CC	Increased risk of diabetes, decreased beta-cell function [ref] [ref] [ref]
HNF4A	rs2144908	A	—	Increased risk of diabetes in conjunction with other variants [ref] [ref]

Details on the genetic variants:

MTNR1B Gene: Melatonin receptor in the pancreas

Melatonin is a signaling molecule that rises in the evening (called dim light melatonin onset) and peaks during the night. Insulin sensitivity is lowest at night, and the melatonin receptors in the pancreatic islets modulate insulin secretion from the beta-cells.[\[ref\]](#)

The MTNR1B gene codes for the melatonin receptor. A genetic variant in MTNR1B (rs10830963 G) was initially identified as a general risk factor for diabetes. But subsequent studies show that the increased risk is mainly for those who eat later at night.[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)

Consider for MTNR1B:

- *Eat dinner earlier:* Simply shifting your dinner to be earlier and not snacking at night has been shown in studies to mitigate the risk from this variant.
- *Don't eat breakfast too early:* Another study found that carriers of the G allele had a longer duration of melatonin production — lasting further into the morning hours (41 minutes). It is possible that getting up early and eating breakfast immediately may not be ideal for this genetic variant.[\[ref\]](#)

SLC30A8 gene: Zinc transporter and beta-cell function

The SLC30A8 gene codes for the zinc transporter ZnT-8. This zinc transporter is found in pancreatic beta-cells and transports the zinc from the cytoplasm into insulin secretory vesicles, stabilizing it and preventing degradation.[\[ref\]](#) The pancreatic beta-cells have the highest levels of zinc of any tissue.[\[ref\]](#)

Consider for SLC30A8:

- *Increase your zinc if it is low:* Research shows that increasing zinc levels by 10 ug/dl decreased the odds of type-2 diabetes for everyone by a little bit, but those with the T/T genotype for rs13266634 had a greater decrease than those who carried the C allele (risk genotype). The majority of studies show that carriers of the rs13266634 C allele are at an increased risk of diabetes, but this may not be true for all population groups. The risk may depend on the normal dietary intake of zinc in the populations that are being studied.[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)
- *Zinc-rich foods include:* oysters, beef, crab, pork, beans (soaked first), pumpkin seeds, and chicken.[\[ref\]](#)

IRS1 gene: Insulin Resistance and higher insulin production

The IRS1 gene codes for a key protein in the insulin-stimulated signal pathway.[\[ref\]](#) The genetic variants of this gene are associated with insulin resistance and hyperinsulinemia rather than impaired beta-cell function.[\[ref\]](#)

Consider for IRS1:

- *Vitamin D:* Vitamin D increases insulin sensitivity for the insulin receptor and stimulates insulin release. Carriers of the rs2943641 T/T genotype had an even greater reduction in the risk of diabetes with higher levels of vitamin D.[\[ref\]](#)[\[ref\]](#) The only way to know your vitamin D level is to get a blood test. If you are on the low end, either expose your skin to the sunshine between 10:00 am – 2:00 pm or consider supplementing with vitamin D to raise your levels to the healthy range.
- *Weight loss diet:* If you need to lose weight, one clinical trial of different types of diets found that a low-fat diet (high in non-refined carbs with fiber) worked best for people with the IRS1 rs2943641 C/C genotype, but not for the C/T or T/T genotypes.[\[ref\]](#) Another (small) study found that a low-fat diet worked best for those with the rs2943641 C/T genotype.[\[ref\]](#)

HHEX gene: insulin response to glucose

The HHEX protein interacts with signaling molecules and plays a role in the embryonic development of the liver, thyroid, and pancreas. Variants in the HHEX gene are linked to impaired glucose-stimulated insulin response.[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)

Consider for HHEX:

- *Low glycemic index diet:* Lower insulin secretion in people who carry the HHEX genes makes it important to eat a lower glycemic diet. Vegetables and whole foods generally require less insulin to be released after eating them (compared with processed foods). The key for carriers of the HHEX variants may be to figure out which foods spike glucose levels (via frequent testing or a continuous blood glucose monitor) and avoid those foods.

KCNJ11 gene: Potassium channel in beta cells in the pancreas

The KCNJ11 gene encodes a protein in the pancreas that is directly involved in insulin release. Glucose activates this protein, which releases insulin from the pancreas. The rs5219 T allele gives a decreased insulin response to glucose.[\[ref\]](#)[\[ref\]](#)

Consider for KCNJ11:

- *Cut refined carbs:* Reducing sugar and refined carbs should help people with the KCNJ11 gene variants. Everyone's insulin response to food is somewhat unique, so a continuous glucose monitor or frequently checking your blood glucose level after eating different foods can give you a better idea of which foods to avoid.
- *Decrease alcohol consumption:* Research shows that chronic alcohol consumption combined with the KCNJ11 variant increases the risk of type 2 diabetes.[\[ref\]](#)

KCNQ1 gene: pancreatic islet cell insulin secretion

The KCNQ1 gene encodes a potassium voltage-gated channel KQT-like subfamily, member 1. This protein is essential in the pancreatic islet cells and the regulation of insulin secretion.[\[ref\]](#) Variants here decrease the risk of diabetes (good!).

PPARG gene: Metabolic syndrome and insulin resistance

The PPARG gene codes for a protein important in promoting the expression of genes involved in fat regulation and energy production. PPARG is needed to regulate the storage of fat and regulate insulin resistance.[\[ref\]](#)

The studies on the **rs1801282** variant (found in about 20% of most populations) show conflicting results regarding whether the variant increases or decreases the susceptibility to diabetes. Part of this may be due to dietary differences between the population groups studied, and part may be due to exercise.[\[ref\]](#)[\[ref\]](#)[\[ref\]](#)

Consider for PPARG:

- *Exercise works for increasing glucose tolerance:* People carrying the G allele for rs1801282 had a greater benefit from exercise for increasing glucose tolerance.

Multiple studies show that carriers of the G allele are more responsive to 'beneficial health effects of lifestyle interventions.'[\[ref\]](#)[\[ref\]](#)

PCSK9: Low LDL Cholesterol

Some PCSK9 genetic variants are solidly linked to lifelong, lower LDL cholesterol levels, which generally cuts the risk of heart disease by about 50% (a good thing). But the flip side is that these variants are also linked with an increased risk of type 2 diabetes.[\[ref\]](#)

CDKAL1 gene: Decreased insulin release

The CDKAL1 gene codes for cyclin-dependent kinase 5, which is a protein expressed in pancreatic islet cells.[\[ref\]](#) CDKAL1 genetic variants impact the release of insulin in response to glucose. Not enough insulin released can lead to higher blood glucose levels.

Consider for CDKAL1 variants:

- *Link to medications:* Sulfonylureas, a type of antidiabetic drug, have been shown not to work as well for people with the rs7763992 or rs7754840 risk alleles.[\[ref\]](#)
- *Low glycemic index diet:* Decreased insulin release may mean that a low glycemic-index diet would be helpful here. One study in women with gestational diabetes found that 'healthy lifestyle intervention' was statistically effective for people with the risk allele for this variant.[\[ref\]](#)

Beta Cell Function (TCF7L2 gene):

The **TCF7L2** (transcription factor 7-like 2) gene activates many genes involved in type 2 diabetes, including glucagon-like peptide 1 (GLP1). Genetic variants are associated with a decreased/impaired beta-cell function.[\[ref\]](#)[\[ref\]](#) People with insulin resistance and impaired glucose tolerance also have decreased TCF7L2.[\[ref\]](#)

Consider for TCF7L2:

- *Glycemic Index Matters:* A study found that those with the TCF7L2 variant had a much higher risk of diabetes (over twice the risk) if they had a diet with a high glycemic index. Here is a chart of the glycemic load of common foods: [Glycemic Index Chart](#). Keep in mind that everyone is individual when it comes to how their body reacts to foods, so use the glycemic index charts and cookbooks as more of a *starting point* rather than something that is written in stone for everyone – best way to know for yourself is to check your response to different foods with either a continuous blood glucose monitor or a finger prick test.
- *Dietary fiber:* A large study that divided people by dietary fiber intake found that people carrying the rs12255373 T allele were actually at a higher risk of type 2 diabetes with high fiber intake. Note that the high fiber intake correlates to high whole-grain carbohydrate intake.[\[ref\]](#) Other studies show that people without the TCF7L2 risk alleles are likely to benefit from high-fiber diets for preventing diabetes, but those who carry the risk allele do not benefit from a high-fiber diet.[\[ref\]](#)[\[ref\]](#)
- *Curcumin:* Animal and human cell studies showed that curcumin stimulates TCF7L2, which theoretically would be helpful for people with the TCF7L2 risk alleles.[\[ref\]](#)[\[ref\]](#)

- *Medication choices:* One study found that people carrying a TCF7L2 variant did not respond as well to the class of diabetes medications known as sulfonylureas. This may be something to discuss with your doctor if you are on a diabetes medication that isn't working well for you. The study notes that TCF7L2 variants did not impact metformin response.[\[ref\]](#)

HNF4A gene:

The HNF4A gene encodes a transcription factor that turns on other genes in the liver and the pancreas.

Consider for HNF4A:

- *Luteolin*, a flavonoid found in celery and parsley, decreases HNF4A and inhibits apoB-containing lipoprotein secretion from the liver.[\[ref\]](#)

The information provided here is for educational and informational purposes only. Please consult with your doctor for any medical advice.